Risk Alert Classifier

Part A: Conceptual Understanding (Theory) | Supervised Learning

1. What is Logistic Regression and why is it suitable for classification?

Logistic Regression is a supervised learning algorithm used to predict the probability of a class. It applies the logistic (sigmoid) function to produce a probability between 0 and 1, which can be converted into a class label using a threshold. It is useful for binary tasks such as identifying low-risk and high-risk customers.

2. Explain classification performance metrics and why accuracy alone is insufficient.

Classification metrics measure how well a model predicts class labels. Common metrics include accuracy, precision, recall, F1-score and AUC-ROC. Accuracy alone can be misleading when classes are imbalanced, because a model may predict the majority class correctly while missing many high-risk customers.

3. Define Type-I Error and Type-II Error in the context of risk prediction.

A Type-I Error (false positive) occurs when a low-risk customer is incorrectly classified as high-risk. A Type-II Error (false negative) occurs when a genuinely high-risk customer is classified as low-risk. In banking, false negatives can be especially costly because risky customers may go undetected.

4. Explain Precision, Recall, F1-Score, TPR, and FPR.

Precision is the proportion of predicted high-risk customers who are truly high-risk; Recall is the proportion of actual high-risk customers correctly identified. The F1-score is the harmonic mean of precision and recall. True Positive Rate (TPR) equals recall, while False Positive Rate (FPR) is the proportion of actual low-risk customers incorrectly predicted as high-risk.

5. What is AUC-ROC and how does it help in evaluating classifiers?

The ROC curve plots True Positive Rate against False Positive Rate at different classification thresholds. AUC is the area under this curve and summarizes how well a classifier separates the two classes. A value near 1 indicates strong discrimination, while a value near 0.5 indicates performance similar to random ranking.

6. Why does imbalanced data create problems in classification models?

Imbalanced data occurs when one class has many more examples than another, such as when high-risk customers are rare. A model may favor the majority class and achieve high accuracy while failing to detect minority-class cases. Techniques such as under-sampling, over-sampling, SMOTE and ADASYN can help, and recall and F1-score should be checked alongside accuracy.

End of Part A — Theory